

3.1 Thread Priority And Scheduling

An operating system has to juggle multiple processes, threads, and priorities. Essentially, all the programs running on your computer—be they operating system services or your favourite word processor—need to access a limited set of resources, the most limited and the most important of which is the computer's process itself or the CPU—generally referred to as CPU cycles. These tasks of juggling are handled by the kernel of the OS, and this very act and manner of juggling has changed quite a bit under Vista.

Vista offers a new scheduler for multimedia tasks called the Multimedia Class Scheduler Service (MMCSS), which goes a long way in ensuring that multimedia applications are able to offer smooth and error-free playback to the end-user. Vista also ensures that tasks get their share of CPU cycles in a fair manner. This is especially necessary in this age of multi-core processors, and virtualisation at both the hardware and software levels.

In order to ensure that multiple threads get a go at a common CPU resource, a method of “interrupting” a process is employed at the kernel level. An “interrupt” essentially ensures that multiple processes get a shot at the CPU cycles. In many ways you can imagine the kernel as the personal secretary of the CPU who decides to allow one person to access the CPU every few milliseconds. Of course, things are not this simple. What the kernel does practically is that it “interrupts” whatever is happening every few seconds and then hands over the CPU to another task. This periodic interruption might be unfair to one or more processes: let's continue with the personal secretary analogy. Say you have been given an appointment with the CPU at 10.00 AM, and you arrive there on the dot. Now the kernel might run its interrupt routine at exactly the same time that your meeting was supposed to take place (the kernel does this periodically, so there is no saying which process might get unlucky and when)—thus you can no longer meet with the CPU, since the kernel will count the time it has interrupted against your scheduled meeting; the kernel